

READ ME

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Details on Content

In this project, I developed a workflow to fetch, process, and visualize intraday data (1minute interval) for various financial instruments over a monthly timeframe (30 days) implementing it in Jupyter and Python files as outlined below:

- **fetching_monthly_intraday_data.ipynb:** This Jupyter notebook serves as the core file where I set up the code to retrieve and process intraday data. It also generates output summaries and visualizations for each instrument.

Libraries Used

The following libraries were used for data handling, analysis, and visualization:

- **Pandas:** For managing and analyzing the data.
- **NumPy:** For numerical operations, particularly with arrays.
- **Matplotlib and Seaborn:** For plotting and visualizing the data.
- **yfinance:** For fetching financial data directly from Yahoo Finance.
- **datetime:** To handle date and time data for setting custom intervals and date ranges.

API Restriction

One limitation to note is that the Yahoo Finance API, accessed via the `yfinance` library, restricts intraday data retrieval to the past 30 days (1 month). This restriction required setting intervals within that period to obtain the necessary data. As such, for extended data needs, this monthly limit must be managed by fetching data periodically.

Files (Input and Output)

The code generates the following types of files for every instrument for a given Start and End Date.

These files are exported in the **output_monthly_intraday** folder:

- 1. **Intraday Data Summary Files:** These CSV files provide statistical summaries (e.g., count, mean, standard deviation, min, max) of intraday data for each instrument over the entire date range. A separate csv file is created to combine the statistics for all instruments for a given start and end date
- 2. **Daily Data Files:** These files capture daily price metrics (Open, Close, High, Low) for each instrument from the start to the end date.
- 3. **Graph Files:** These plots visually represent each instrument’s **price trend (Close Price)** with shaded areas indicating high-low ranges. The charts provide insights into price movement patterns over the monthly period.

Example:
Start Date: 30 September 2024 (Inclusive)
End Date: 28 October 2024 (Inclusive)
Instrument: ZN (ticker is ZN=F)

Category	File Name (Example)
Intraday Data Summary Files	Intraday_monthly_summary_ZN_2024-09-30_to_2024-10-28
Intraday Data Summary Files (Combined)	All_tickers_Intraday_monthly_data_summary_2024-09-30_to_2024-10-28
Daily Data Files	Intraday_monthly_data_ZN_2024-09-30_to_2024-10-28
Graph Files	Monthly_intraday_returns_All_Dates_ZN_2024-09-30_to_2024-10-28

These files are exported in **output_monthly_intraday_bps_ticks_ZN** folder:

1. ZN Intraday returns data in (bps and ticks)
2. ZN Intraday returns data summaries (bps and ticks)
3. ZN Intraday returns data graphs (bps and ticks together in one graph)

Tickers Used

The project retrieves data for the following financial instruments:

- **10-Year T-Note Futures** (ZN=F)
- **US Dollar Index** (DX-Y.NYB)
- **Crude Oil Futures** (CL=F)
- **Gold Futures** (GC=F)
- **Nasdaq 100 Futures** (NQ=F)
- **Dow Jones Industrial Average** (^DJI)
- **S&P 500** (^GSPC)

Each ticker corresponds to a unique financial instrument, and I utilized them to explore diverse data points within the financial market.

By using this setup, I achieved a streamlined method for retrieving, processing, and visualizing monthly intraday data. This project provides a structured, reproducible approach for analyzing financial data and can easily be adapted for other financial instruments by modifying the tickers and date ranges in the code.